Hamed Hojatian

Ph.D. Candidate

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Education

Fall 2019 – Now
Polytechnique Montréal (PolyMTL)
Ph.D. in AI-aided wireless communication systems
Supervisors: Prof. François Leduc-Primeau, Prof. Jean-François Frigon
GPA: 4.0/4.0

2014-2017
Isfahan University of Technology (IUT)
M.Sc. in communication system engineering
Supervisor: Prof. Mohammad Javad Omidi
GPA: 3.7/4.0

2010-2014
Babol (Noshirvani) University of Technology (NIT)
B.Sc. in electrical engineering
Supervisor: Prof. Hossein Miar-Naeimi
GPA: 3.1/4.0

Research skills and Qualifications

- Strong proficiency in machine learning techniques such as regression, classification, clustering, and neural networks;
- Skilled in developing and implementing machine learning algorithms using a range of techniques, including supervised learning, unsupervised learning, and reinforcement learning;
- Understanding of deep reinforcement learning techniques, with experience implementing and optimizing various algorithms such as Q-learning, and deep deterministic policy gradients (DDPG);
- Experience with massive MIMO, beamforming techniques, coordinated beamforming, distributed antennas, cell-free communication and their applications in wireless communication systems, 5G beyond, and 6G;
- Experience in developing and implementing deep learning-based beamforming solutions, with a focus on optimizing beamforming performance for wireless communication systems;
- Skilled in using quantization techniques to optimize deep neural networks (DNNs) for specific hardware constraints and other use cases;
- Developed and implemented machine learning models for wireless communication systems, including channel estimation, interference cancellation, and beamforming;
- Published research papers in leading academic journals (IEEE) and presented research findings at international conferences;
- Collaborated with cross-functional teams to identify research opportunities and develop solutions that improved the reliability, efficiency, and scalability of wireless communication systems;
- Mentored graduate and undergraduate students on research projects and provided guidance on technical skills and research methodologies;

Skills

Tools and Languages | Python, Pytorch, C++ (intermediate), MATLAB, and Microsoft Office.

Languages | English, French (basic), Persian (native)

Research Experience

Ph.D. ***AI-aided Beamforming in Massive MIMO Systems***

- Designing hybrid beamforming techniques for wireless networks, including an innovative RSSI-based hybrid beamforming design that leverages deep learning algorithms to enhance spectral and energy efficiency;
- Developing unsupervised learning techniques for intelligent beamforming and coordinated hybrid beamforming in cell-free massive MIMO systems;
- Exploring subarray hybrid beamforming and efficient hardware design driven by unsupervised learning techniques, to reduce the complexity of wireless communication systems;
- Publishing in top-tier conferences and journals in the field of wireless communications, my research showcases my expertise in developing cutting-edge technologies to improve wireless network performance;
- Filing a patent for the invention in collaboration with Ericsson as a co-op research intern, I developed a novel AI-aided method for efficient massive MIMO beamforming.

Work Experience

Nov 2022 - present **Ericsson**
Research intern, Co-op research

Mar 2019 - Aug 2019 **Polytechnique Montréal**
Research Assistant

Nov 2015 - Nov 2018 **Mobile Telecommunication Company of Iran (MCI)**
Radio planning and optimization engineer

- Live network Ericsson OSS optimization and parameter tuning for 2G, 3G, and LTE;
- Network Planning (2G, 3G, and LTE), new site baseline parameters tuning and planning;
- Project manager of RNC rehoming and RNC planning.

Publications

2023
1. **Hojatian, H.**, Mlika, Z., Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Learning Energy-Efficient Transmitter Configurations for Massive MIMO Beamforming**". *submitted* ***IEEE Transactions on Machine Learning in Communications and Networking*** (2023).

2022
2. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Decentralized Beamforming for Cell-Free Massive MIMO with Unsupervised Learning**". ***IEEE Communications Letters*** (2022).
3. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Flexible Unsupervised Learning for Massive MIMO Subarray Hybrid Beamforming**". ***IEEE GLOBECOM*** (2022).

2021
4. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Unsupervised Deep Learning for Massive MIMO Hybrid Beamforming**". ***IEEE Transactions on Wireless Communications*** (2021).

2020
5. **Hojatian, H.**, Nguyen Ha, V., Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**RSSI-Based Hybrid Beamforming Design with Deep Learning**". ***IEEE ICC*** (2020).

2016
6. **Hojatian, H.**, Omidi, M. J., Saeedi-Sourck, H. & Farhang, A. "**Joint CFO and Channel Estimation in OFDM-based Massive MIMO Systems**". ***IEEE IST*** (2016).

Patent

2023 H. Hojatian, Z. Mlika, J. Nadal, J. Frigon and F. Leduc–Primeau, "**Learning Energy Efficient Massive MIMO Beamforming**", U.S. patent, filed by Ericsson, PCT/IB2023/053214.

Mentoring

Throughout my Ph.D. studies, I had the privilege of mentoring a master student and an internship student in our research project on wireless communication systems. Currently, I am mentoring a Ph.D. student, providing guidance and support in their research work. Through my mentorship, I have helped these students build their technical skills and gain valuable research experience.

Community Service

IEEE Journals reviewer: Selected Areas in Communications (**J-SAC**), Transactions on Wireless Communications (**TWC**), Transactions on Communications (**TCOMM**), Wireless Communications Letters (**WCL**), Communications Letters (**CommL**), and Transactions on Vehicular Technology (**TVT**). Details in here.

Talks and Teaching Experience

2022 Paper Presentation, **IEEE GLOBECOM 2022**, Rio-de-Janeiro, Brazil.
Hybrid-Seminars on Selected Topics in Signal Processing.
Presenting my research progress in **IVADO - Digital October.**

2021 Presenting my research progress in **IVADO - Digital October.**

2020 Paper Presentation, **IEEE ICC 2020**, Dublin, Ireland.
Presenting my research topic in **IVADO - Digital October.**

2013 Teaching assistant for **Electronic I**.
Teaching assistant for **Electronic circuit I**.

2012 Teaching assistant for **Digital Signal Processing**.

Certificates

2022 Introduction to Unix Shell, **Calcul Quebec**.
Massive data analysis with Spark, **Calcul Quebec**.
Introduction to GPU Programming with Python, **Calcul Quebec**.

2019 4th **IVADO/MILA** SCHOOL IN DEEP LEARNING.

2018 DEEP LEARNING WITH PYTORCH, **Udemy**.
LTE OPTIMIZATION OF RADIO NETWROK (3GPP), **TechCom**.

2017 LTE COUNTER AND KPI (ERICSSON), **TechCom**.
LTE RADIO NETWORK PLANNING, **TechCom**.

Strengths

Hard-working, Decision Making, Ability to Work Under Pressure, Motivator & Leader Conflict Resolution, Time Management, Target oriented, Flexibility, and Adaptability